

EXPERIMENT 6

Explainability for Image Data using XRAI

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Batch	C
Experiment	6
Subject	Explainable AI / XAI

Repository

<https://github.com/CodeCraftsmanRaj/xAI/tree/main/Exp6>

1. Aim

To understand the concept of Explainable AI (XAI) and implement the XRAI (eXplanation with Ranked Area Integrals) attribution method to visually interpret the predictions of a deep Convolutional Neural Network (CNN) image classifier.

2. Theory / Methodology

The experiment follows the XRAI methodology described in the laboratory instructions. The image is first over-segmented into coherent regions using Felzenszwalb graph-based segmentation at multiple granularities. Integrated Gradients (IG) is then used to obtain pixel-level attribution for the target class, with multiple baselines such as black and white images. Attribution is aggregated within candidate regions, the regions are greedily ranked by their incremental contribution, and the ranked regions are converted into an XRAI heatmap. Higher-ranked regions therefore indicate image areas that contribute more strongly to the model's prediction.

For this implementation, the XRAI methodology was implemented natively in PyTorch rather than through the TensorFlow saliency API. A pretrained InceptionV3 model with ImageNet weights was used. Images were resized to 299×299 RGB inputs and ImageNet normalization was applied explicitly.

3. Experimental Setup

Component	Configuration
Model	InceptionV3, pretrained on ImageNet
Framework	PyTorch
Input	RGB, 299 × 299
Attribution	Integrated Gradients
Segmentation	Felzenszwalb graph-based segmentation
XRAI	Region aggregation + greedy ranking
Baselines	Black and white images
Hardware	NVIDIA GeForce RTX 4050 Laptop GPU
Execution device	CUDA

4. Prediction Results

The experiment was run on four sample images: astronaut, coffee, chelsea, and rocket. The following values are taken from the generated experiment results.

Image	Predicted class	Confidence
astronaut	football helmet	~40.15%
coffee	espresso	~98.51%
chelsea	Egyptian cat	~93.81%
rocket	crane	~46.00%

5. Top-5 ImageNet Predictions

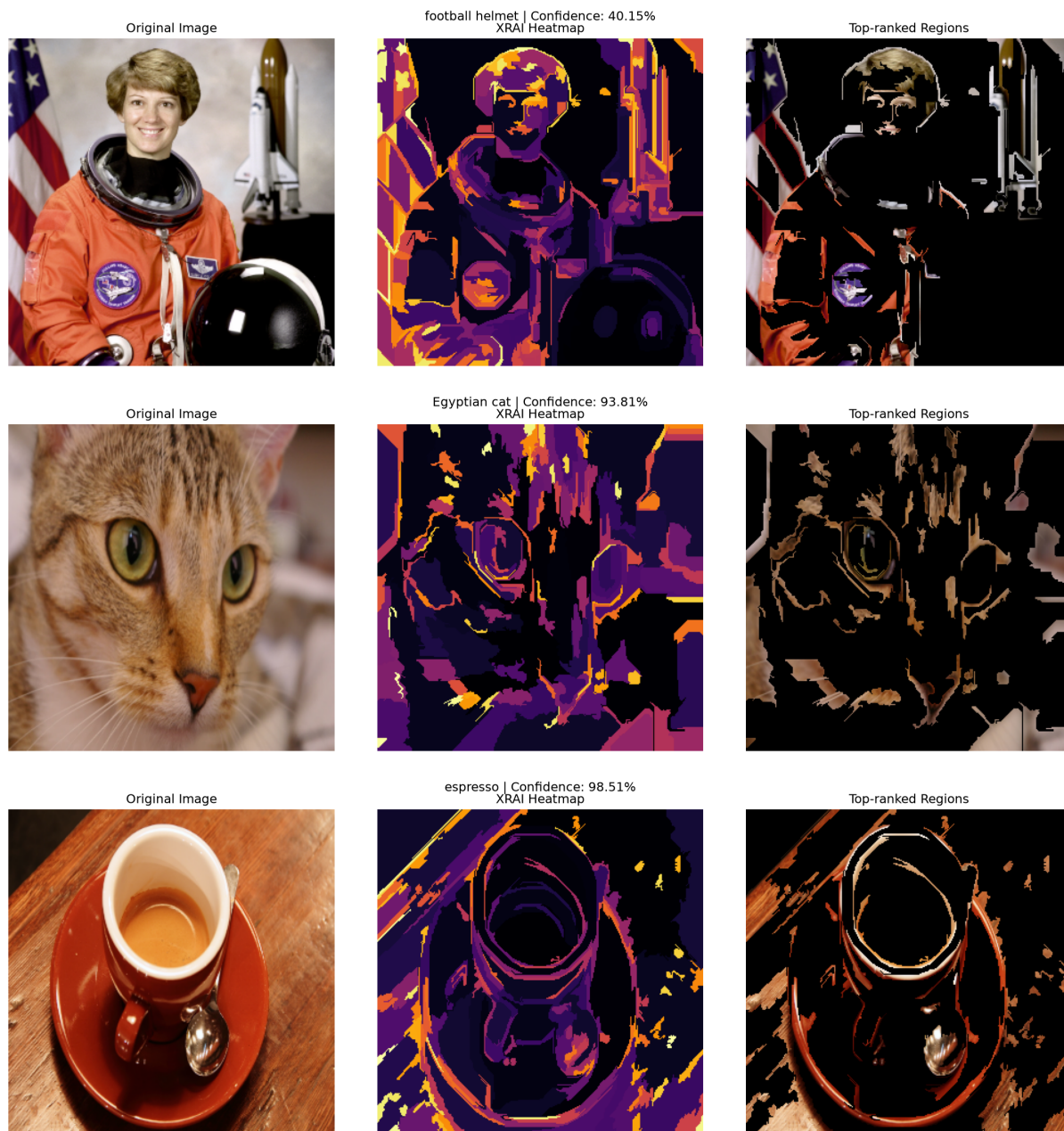
The generated predictions show how ImageNet classification can differ from the everyday semantic identity of an image. For example, the astronaut image was classified primarily as a football helmet, while the coffee image was classified very confidently as espresso. The chelsea image was classified as

Egyptian cat, and the rocket image as crane. These outputs provide the target classes whose visual evidence is subsequently examined by XRAI.

6. XRAI Visualization Results

The XRAI pipeline produces three principal visual outputs for each image: the original image, the raw XRAI attribution heatmap, and a visualization/mask emphasizing the top-ranked influential regions. The generated result set contains these visualizations. Candidate-region counts reported by the experiment include 247 regions for astronaut and 108 regions for rocket, illustrating that segmentation depends on image content and boundaries.

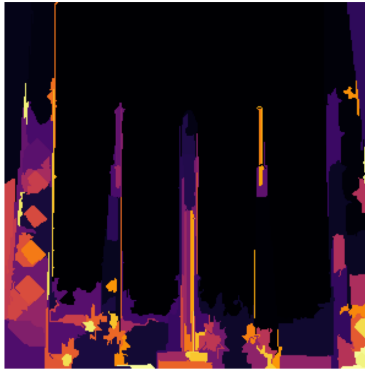
Generated Visualizations



Original Image



crane | Confidence: 46.00%
XRAI Heatmap



Top-ranked Regions



7. Task 1 — Prior Hypothesis vs XRAI

Before computing XRAI, the task requires a prediction of which image region the model will rely on and which region it will likely ignore. The hypothesis should then be compared with the revealed XRAI heatmap. This provides a direct check of whether human expectation agrees with the visual evidence actually used by the classifier.

For the astronaut image, the classifier's target class was football helmet. Therefore, the comparison should be made against the XRAI regions supporting that predicted class, rather than assuming that the model must focus on the semantically intended concept 'astronaut'. This distinction is important because explainability methods explain the model's prediction, including cases where the prediction itself is unexpected.

8. Task 2 — Patch Sensitivity Analysis

The patch experiment masks an image region with a gray patch and measures the change in the target-class confidence. The generated patch-search CSV records the patch location, patch dimensions, confidence, and confidence drop. A large confidence drop indicates that the masked area contained evidence important to the original prediction.

The recorded patch search contains 4360 evaluated patches. The largest recorded confidence drop was 0.401075, at location (160, 180) with patch size 90×90 .

9. Interpretation of Results

The results demonstrate the complementary roles of XRAI and patch masking. XRAI converts pixel-level Integrated Gradients into ranked spatial regions, making it easier to see where the classifier obtains evidence for its target class. Patch masking provides a separate sensitivity check: when masking an important region causes a substantial confidence drop, the observation can be compared with the corresponding XRAI attribution. Agreement between the two analyses provides stronger evidence that the region is relevant, while disagreement can reveal sensitivity to patch size, stride, segmentation boundaries, or other model behavior.

10. Conclusion

The experiment successfully applies the XRAI explainability methodology to image classification using a PyTorch-based implementation. InceptionV3 generated ImageNet predictions for four sample images, Integrated Gradients supplied pixel-level attribution, Felzenszwalb segmentation supplied candidate regions, and greedy region ranking produced the XRAI explanations. The patch-sensitivity experiment additionally examined how masking local image regions changes model confidence. Together, these steps demonstrate how XAI can expose the visual evidence behind CNN predictions, including cases where the model's prediction differs from the expected semantic label.

11. Repository

The source code, configuration, experiment scripts, and result artifacts are maintained in:

<https://github.com/CodeCraftsmanRaj/xAI/tree/main/Exp6>